PS-3: Universal File Type Converter Web Application (Beginner)

Team Size: 2–4 Members

1. Background / Introduction

Format incompatibility is a universal frustration. Whether it is a student needing to edit a locked PDF, a designer needing to compress a JPG to a PNG, or a developer extracting text from a document, the need to convert files is constant. While many online converters exist, they are often riddled with intrusive ads, restrictive paywalls, or questionable data privacy policies. This project challenges you to build a clean, efficient, and secure alternative: a dedicated web application that handles multiple file format conversions seamlessly while prioritizing user experience and data privacy.

2. Problem Statement / Objective

Design and develop a full-stack web application that enables users to convert files between various formats (e.g., PDF to DOCX, JPG to PNG, DOCX to TXT). The application must feature an intuitive drag-and-drop interface, provide real-time status updates during the conversion process, and implement strict server-side security measures to ensure uploaded files are safely processed and subsequently deleted.

3. Scope of Work / Core Requirements

- **Frontend Interface:** Develop a responsive, clean UI featuring a drag-and-drop upload zone.
- **Real-Time Feedback:** Implement progress indicators that update the user on the current state of their request (Uploading -¿ Processing -¿ Ready for Download).
- **Multi-Format Processing:** Integrate server-side logic, libraries, or external APIs capable of handling conversions for documents (PDF, DOCX, TXT) and images (JPG, PNG).
- **Secure File Handling:** Develop a backend cron job or automated script that strictly deletes all uploaded and converted files from the server after a set period (e.g., 30 minutes) to protect user data.
- **User Authentication:** Allow users to create accounts (standard email/password) to access a personalized ”Conversion History” dashboard.

4. Technical Constraints / Guidelines

- **File Size Limits:** The backend must enforce a strict maximum file size limit for uploads to prevent server overload.
- **Security:** Uploaded files must be stored in secure, temporary directories and absolutely must not be indexed or publicly accessible prior to conversion.
- **Libraries:** Teams are encouraged to use open-source CLI tools or libraries (like ImageMagick or Pandoc) integrated into their backend, or free-tier conversion APIs.

5. Deliverables

Mandatory Deliverables

- Source code repository (GitHub) containing both frontend and backend code.
- A comprehensive README explaining the supported file formats, the libraries used for conversion, and instructions for local setup.
- A functional web application hosted online

Optional Deliverables

- A demo video showcasing a document conversion, an image conversion, and the user history dashboard.

6. Evaluation Criteria

- **Functionality & Accuracy:** Do the converted files maintain their integrity and formatting?
- **UI/UX & Responsiveness:** Is the drag-and-drop interface smooth? Does the app work well on mobile browsers?
- **Security & Optimization:** Is the file deletion logic robust? Are errors (like unsupported file types) handled gracefully without crashing the app?
- **Code Quality:** Clean separation between the frontend upload logic and the backend processing streams.

7. Selection Criteria

Teams may be shortlisted based on:

- **Project Proposal:** A brief explanation of which formats they plan to support and the specific libraries/APIs they intend to use to achieve the conversions.

8. Recommended Skillsets / Prerequisites (Optional)

- Frontend Frameworks (React, Vue, or Vanilla HTML/JS/CSS)
- Backend Frameworks (Node.js/Express, Python Flask/Django)
- Experience with file handling (multer for Node.js, werkzeug for Flask)
- Basic understanding of server-side scripting.

9. Bonus Extensions (Optional)

- **Bulk Conversion:** Allow users to upload multiple files at once and download the converted results as a single .zip file.
- **OCR Integration:** Implement Optical Character Recognition (OCR) to allow users to extract editable text from flattened images or scanned PDFs.
- **Offline Mode (PWA):** Use WebAssembly (WASM) to perform simple image conversions directly in the browser, without needing to upload the file to a server.

10. Resources / References

- **File Handling:** Documentation on Node.js fs module or Python os module.
- **Conversion Tools:** ImageMagick documentation, Pandoc documentation.
- **APIs:** CloudConvert API (Free Tier).